A SAI MOULI

Backend Software Engineer

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EXPERIENCE

Software Engineer | NatWest Group, Bengaluru — July 2025 – Present

- Cut dashboard response time on large queries from **~3 min to under 30 s** by reworking the backend APIs and data-processing path.
- Built a natural-language-to-SQL engine over a **136-table** reporting database: it turns a plain-English question into SQL, validates it against the schema before it runs, executes it, and returns the answer with a chart.
- Shipped it as a chatbot in the analytics dashboard; stakeholders pull ad-hoc reports without an engineering ticket.

Backend Developer Intern | Aamara Technologies, Bengaluru — March – August 2024

- Built the backend of Techno Rings' shop-floor operations system, used by **20,000 people** (Django REST Framework, MySQL, React): worker–machine assignment, shift logging, tool-life tracking, shift reports.
- Implemented the payroll exception paths as model-level rules: a breakdown pro-rates shift and target, the worker moves to another machine's job, and a day with every machine already staffed records no wage.
- Moved pay rates and thresholds into a config table and enforced assignment uniqueness with DB constraints; daily processing fell from **about four hours to under thirty minutes**.

Software Development Intern | Pods Studio, Bengaluru — Nov – Dec 2023

- Replaced the manual customer-email send step with event-driven Celery tasks and cron jobs; retention improved **27%**.
- Reproduced, root-caused and fixed customer-reported failed in-app payments, owning each from report to patch.

OPEN SOURCE

PrefectHQ/fastmcp | Python framework for MCP servers and clients — 2 PRs merged

- **#4758 (merged):** fixed a lost-update race where two tasks read-modify-wrote one stream's event list; a five-concurrent-write repro kept 1 of 5 IDs. Fixed the race, not the `FileNotFoundError` it surfaced.
- After maintainer review, replaced the store-wide lock with **64 striped asyncio locks** keyed by stream ID, so unrelated sessions don't serialize; the regression test forces the interleave.
- **#4705 (merged):** the response cache stored `is_error=True` results, replaying a transient 503 for the full one-hour TTL. Added a no-cache guard for error results, with retry-path tests.

chroma-core/chroma | open-source vector database — 1 issue, 1 open PR

- **Issue #7560:** showed the CI lint job could never fail: the step set `continue-on-error` and hooks ran under `bash -e`, so the first failure skipped the rest; black, flake8, prettier never ran.
- **PR #7559 (open):** proposes a fix for two undefined-name (F821) defects that step hid, a missing import raising `NameError` under `typing.get_type_hints()`; with a regression test.

PROJECTS

XTXT | plain-text document format | Go reference parser + SDKs — github.com/SaiMouli3/xtxt

- Structure is explicit syntax, not inferred from prose: `@name` directives carry images, tables, charts and typed records, and `xtxt extract` returns them as JSON. Files stay plain text and diff cleanly in Git.
- Wrote the spec, a Go parser and CLI, and a 12-case conformance suite (expected AST + diagnostics) that the Python, JS, Rust, Java and C SDKs run in CI. Published to PyPI and npm; VS Code extension and MCP server.

Autonomous Code Agent | LangGraph, MCP, Tree-sitter, Docker — github.com/SaiMouli3/autonomous-code-agent

- Given a repo URL and a task, runs a LangGraph ReAct agent in a fresh Docker container per run; it edits, runs the tests, then branches, commits, pushes and opens a PR through GitHub's MCP server.
- Built its MCP tool server: file access is path-checked to the workspace, shell runs only in the container, and code search ranks Tree-sitter function/class chunks by embedding similarity, returning file:line ranges.

TECHNICAL SKILLS

Languages: Python, C++, SQL
Backend: FastAPI, Django, Django REST Framework, Flask, REST APIs, WebSockets, Celery
Data & Infra: PostgreSQL, MySQL, Redis, AWS, Docker, Kubernetes, Linux, Git, GitHub Actions
AI Engineering: LangGraph, LangChain, MCP servers, RAG pipelines, embeddings, FAISS
Practices: asyncio concurrency, pytest regression tests, API design, performance optimization

EDUCATION

Ramaiah Institute of Technology | B.Tech, Computer Science and Engineering | CGPA 9.06/10 — 2021 – 2025